GATE ALL BRANCHES



Probability and Statistics

-ENGINEERING MATHEMATICS

Lecture No.- 06



By- Rahul sir

Recap of Previous Lecture



Topic Fundamentals of probability

Topics to be Covered



Topic Classification of events — A, B, C

Topic Conditional probability →

Topic Bayes theorem →

problem solving

Rahul Sir PW

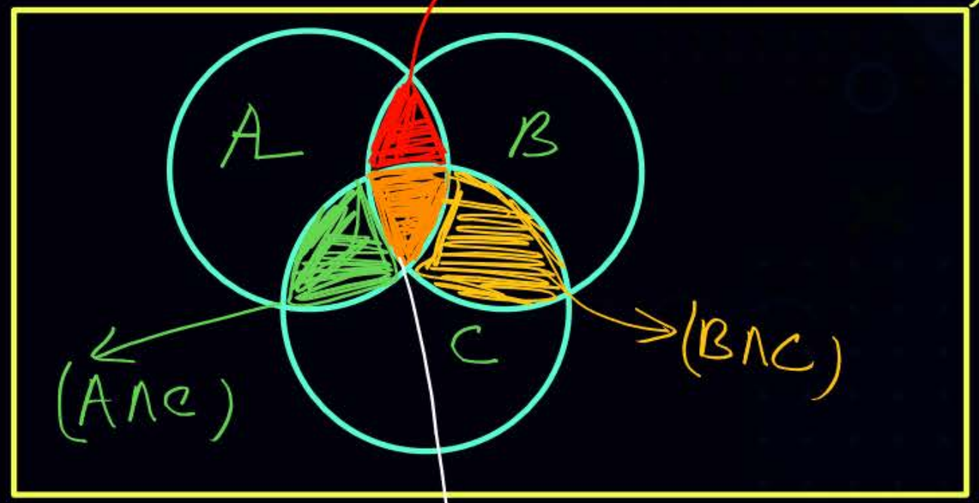
Classification of events: A, B, C (Independent events but something is common)

'something is common'

P[either A or B or C]

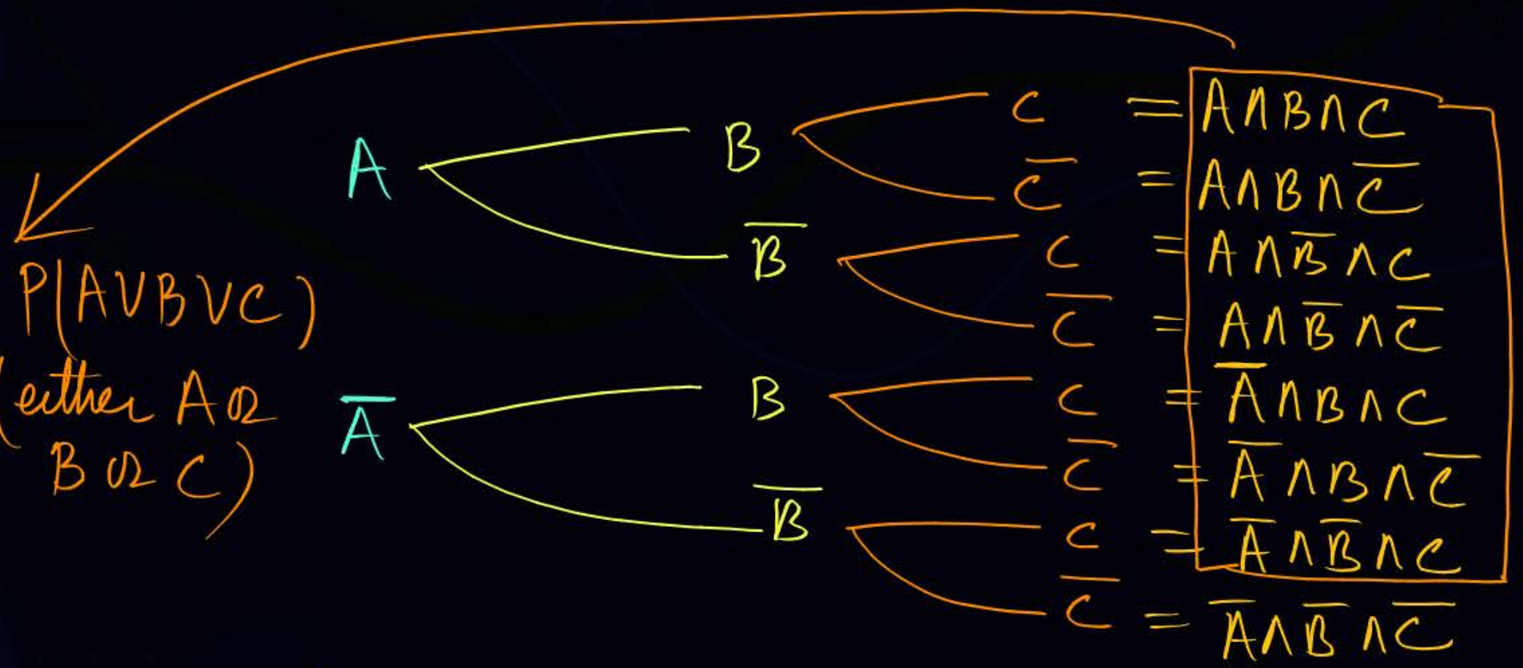
= P(A ∪ B ∪ C)

Using Venn Diagram

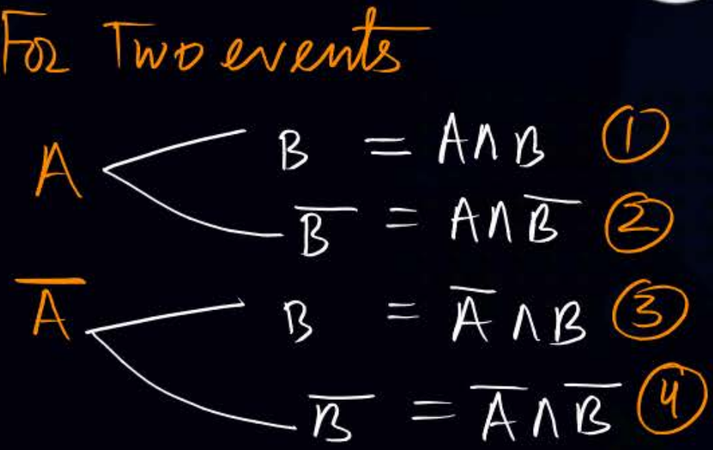


$$P(A \cup B \cup C) = \underset{(8)}{} \underset{(1)}{P(A)} + \underset{(2)}{P(B)} + \underset{(3)}{P(C)} - \underset{(4)}{P(A \cap B)} - \underset{(5)}{P(B \cap C)} - \underset{(6)}{P(C \cap A)} + \underset{(7)}{P(A \cap B \cap C)}$$

$$P(A \cup B \cup C) = P(\text{only } A) + P(\text{only } B) + P(\text{only } C) - P(\text{only } A \text{ and } B) - P(B \text{ and } C) - P(C \text{ and } A) + P(A \text{ and } B \text{ and } C)$$



For Two events



2^2 = 4 events

1) common region $A \cap B \cap C$
2) $A \cap B \cap \bar{C}$ → only A and B
3) $A \cap \bar{B} \cap C$ → only A and C
4) $A \cap \bar{B} \cap \bar{C}$ → only A
5) $\bar{A} \cap B \cap C$ → only B and C
6) $\bar{A} \cap B \cap \bar{C}$ → only B.
7) $\bar{A} \cap \bar{B} \cap C$ → only C
8) $\bar{A} \cap \bar{B} \cap \bar{C}$ → Neither A Nor B Nor C

What is the Prob either 2 or 3 or 5

$P(2 \cup 3 \cup 5)$

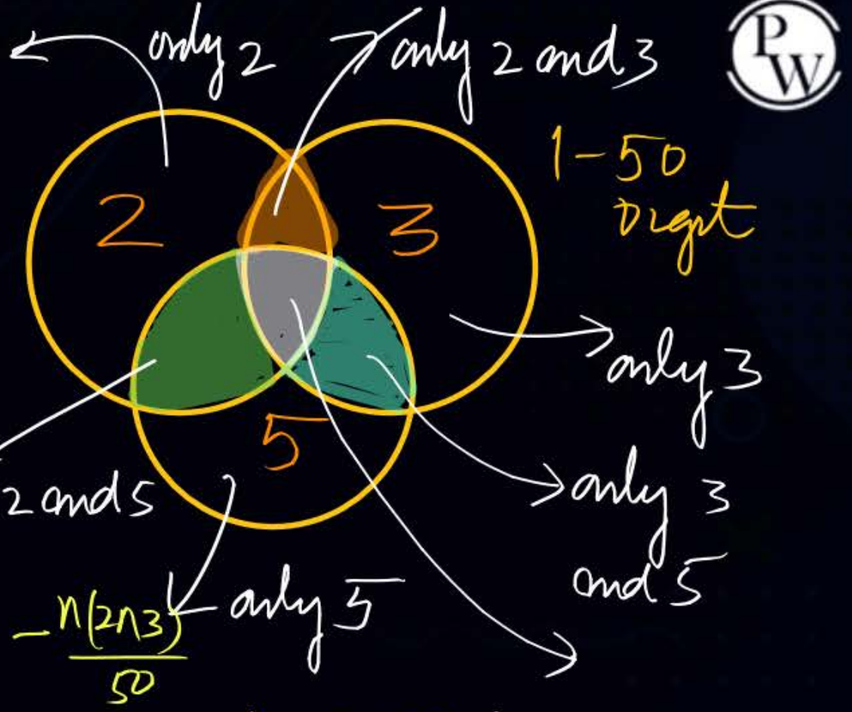
$$P(2 \cup 3 \cup 5) = P(2) + P(3) + P(5) - P(2 \cap 5) - P(3 \cap 5) - P(2 \cap 3) + P(2 \cap 3 \cap 5)$$

$$= \frac{n(2)}{50} + \frac{n(3)}{50} + \frac{n(5)}{50} - \frac{n(2 \cap 5)}{50} - \frac{n(3 \cap 5)}{50} - \frac{n(2 \cap 3)}{50} + \frac{n(2 \cap 3 \cap 5)}{50}$$

$$= \frac{25 + 16 + 10 - 5 - 3 - 8 + 1}{50}$$

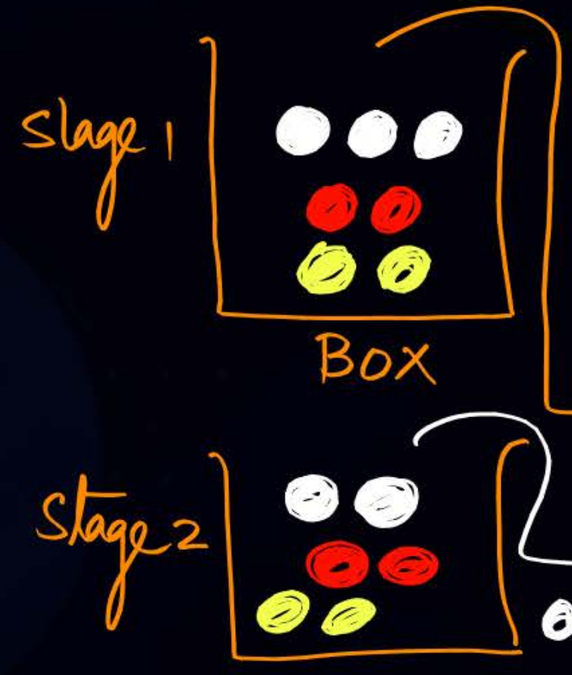
$$= \frac{51 - 16 + 1}{50} = \frac{52 - 16}{50}$$

$$= \frac{36}{50} = \frac{18}{25}$$



Conditional Prob :- Relative Frequency / Brownian moment

(to given a condition)

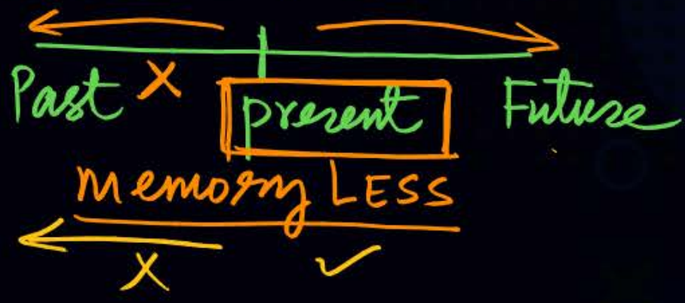


"What is The Probability P(2 white ball Are drawn at random one at a time) (without Replacement)

- First White ball $P(W_1) = \frac{3}{7}$
- SECOND White ball $P\left(\frac{W_2}{W_1}\right) = \frac{2}{6}$

W_1 is already occured

$\frac{W_2}{W_1}$ is occuring.



$$P(W_1 \cap W_2) = P(W_1)\,P\left(\frac{W_2}{W_1}\right)$$

Conditional Prob.

$$P\left(\frac{W_2}{W_1}\right) = \frac{P(W_1 \cap W_2)}{P(W_1)}$$

Conditional Prob.

Conditional Prob.

$P(W_1) \neq 0$

$$P\left(\frac{W_2}{W_1}\right) = \frac{P(W_1 \cap W_2)}{P(W_1)} = \frac{\text{Common region}}{P(W_1)}$$

$P\left(\frac{W_1}{W_2}\right)$ W_2 is already occured and $\frac{W_1}{W_2}$ is occurring.

$$\Rightarrow P\left(\frac{W_1}{W_2}\right) = \frac{P(W_1 \cap W_2)}{P(W_2)} \quad P(W_2) \neq 0$$

Conditional prob

If $W_1 = A$ $W_2 = B$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} \quad P(A) \neq 0$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \quad P(B) \neq 0$$

Conditional Prob.



Topic : Probability and Statistics

#Q. A bag contains 4 ticket numbers (1,2,3,4) and another bag contains 6 ticket numbers (2,4,6,7,8,9). One bag is chosen and ticket is drawn. Then probability that the ticket bears the number 4 is____.

1, 2, 3, (4)

2, 4, 6, 7
8, 9

Box 1

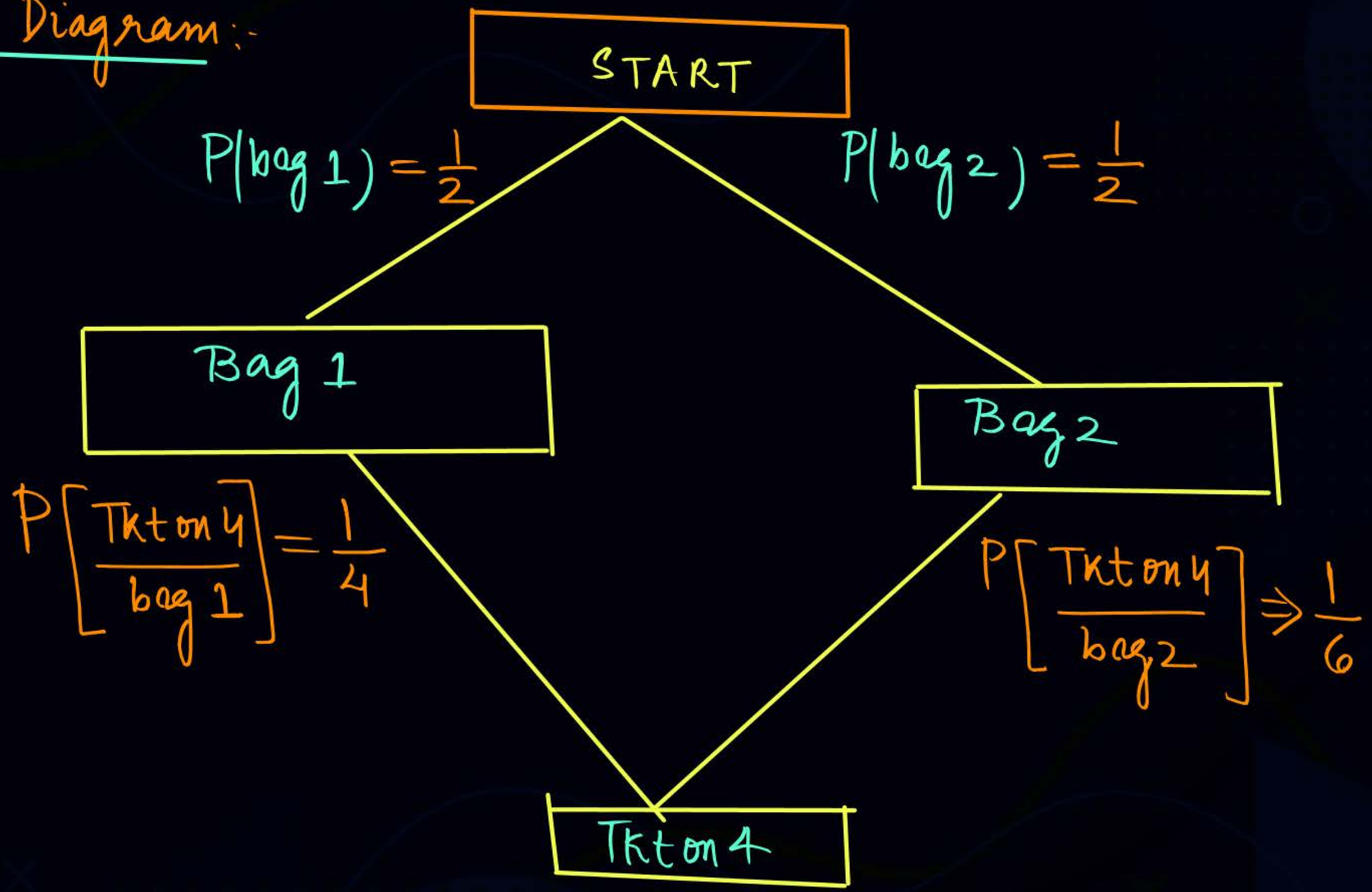
$\frac{1}{2} \times \frac{1}{4}$ +

Box 2

$\frac{1}{2} \times \frac{1}{6}$

$= \frac{5}{24}$

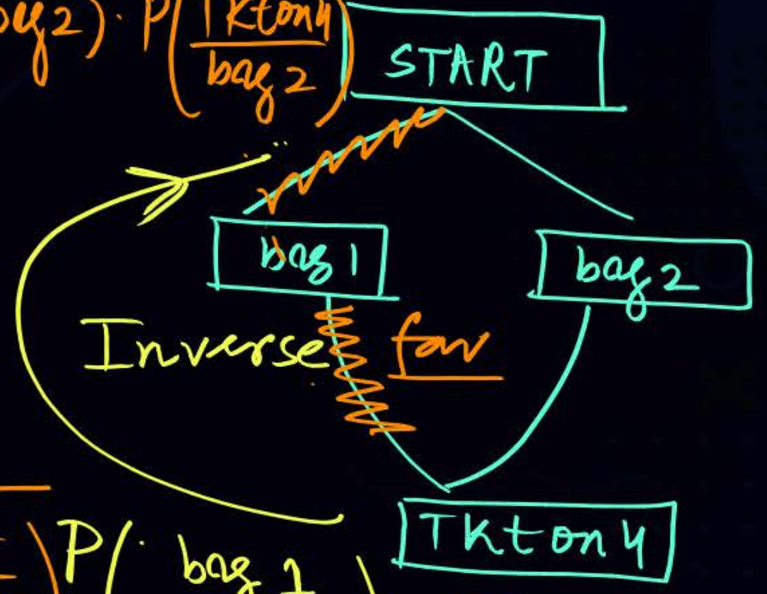
Working Diagram :-



$$P(\text{Tkt on 4}) = P(\text{bag 1}) \cdot P\left(\frac{\text{Tkt on 4}}{\text{bag 1}}\right) + P(\text{bag 2}) \cdot P\left(\frac{\text{Tkt on 4}}{\text{bag 2}}\right)$$

$$= \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{5}{24} \checkmark$$



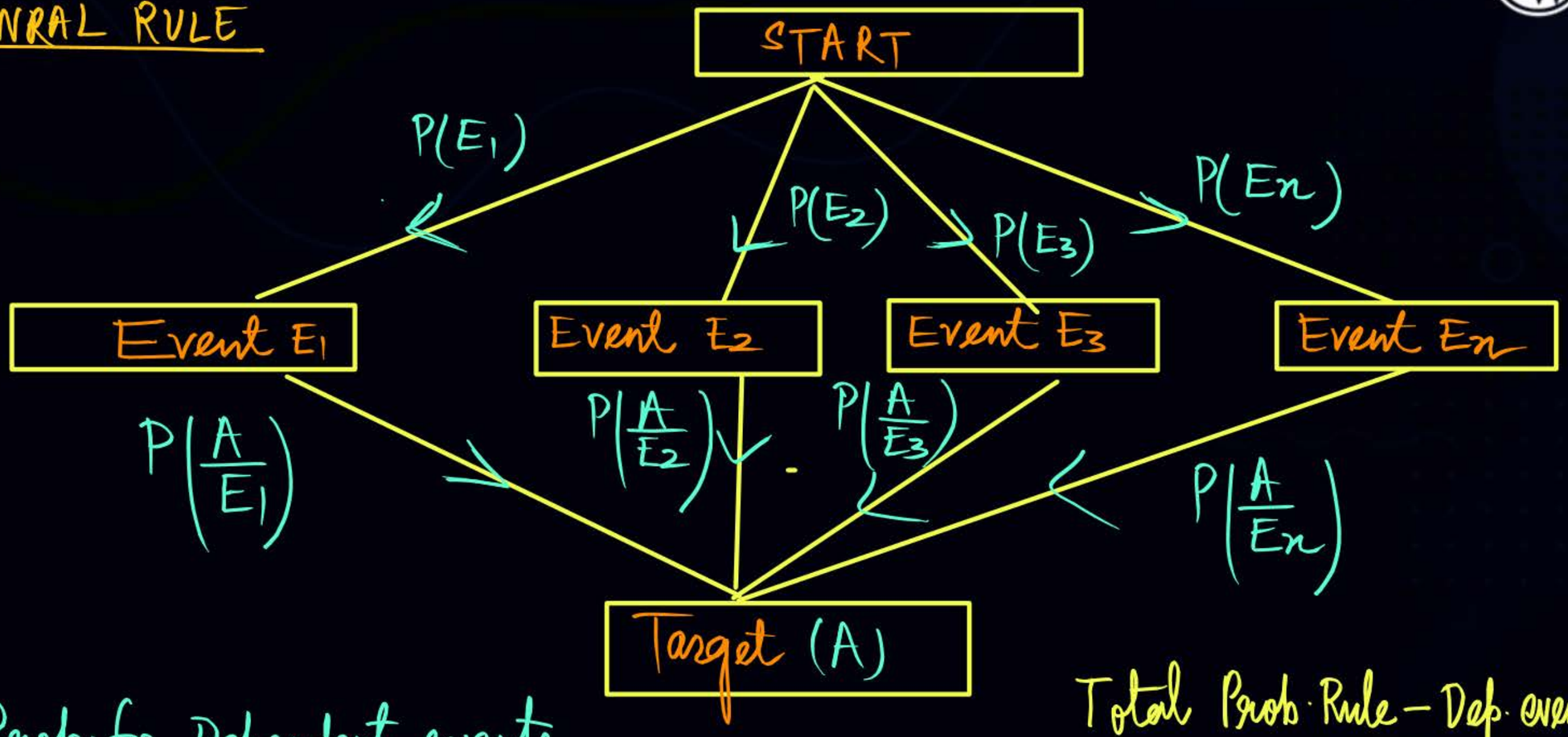
Inverse prob: bayes Theorem

$$P\left(\frac{\text{bag 1}}{\text{Tkt on 4}}\right) \Rightarrow \frac{P(\text{bag 1})\, P\left(\frac{\text{Tkt on 4}}{\text{bag 1}}\right)}{P(\text{bag 1})\, P\left(\frac{\text{Tkt on 4}}{\text{bag 1}}\right) + P(\text{bag 2})\, P\left(\frac{\text{Tkt on 4}}{\text{bag 2}}\right)} P\left(\frac{\text{bag 1}}{\text{Tkt on 4}}\right)$$

$$P\left(\frac{\text{bag 1}}{\text{Tkt on 4}}\right) = \frac{\frac{1}{2} \times \frac{1}{4}}{5/24} = \frac{3}{5}$$

$$P\left(\frac{\text{bag 2}}{\text{Tkt on 4}}\right) = \frac{\frac{1}{2} \times \frac{1}{6}}{5/24} = \frac{2}{5}$$

GENRAL RULE

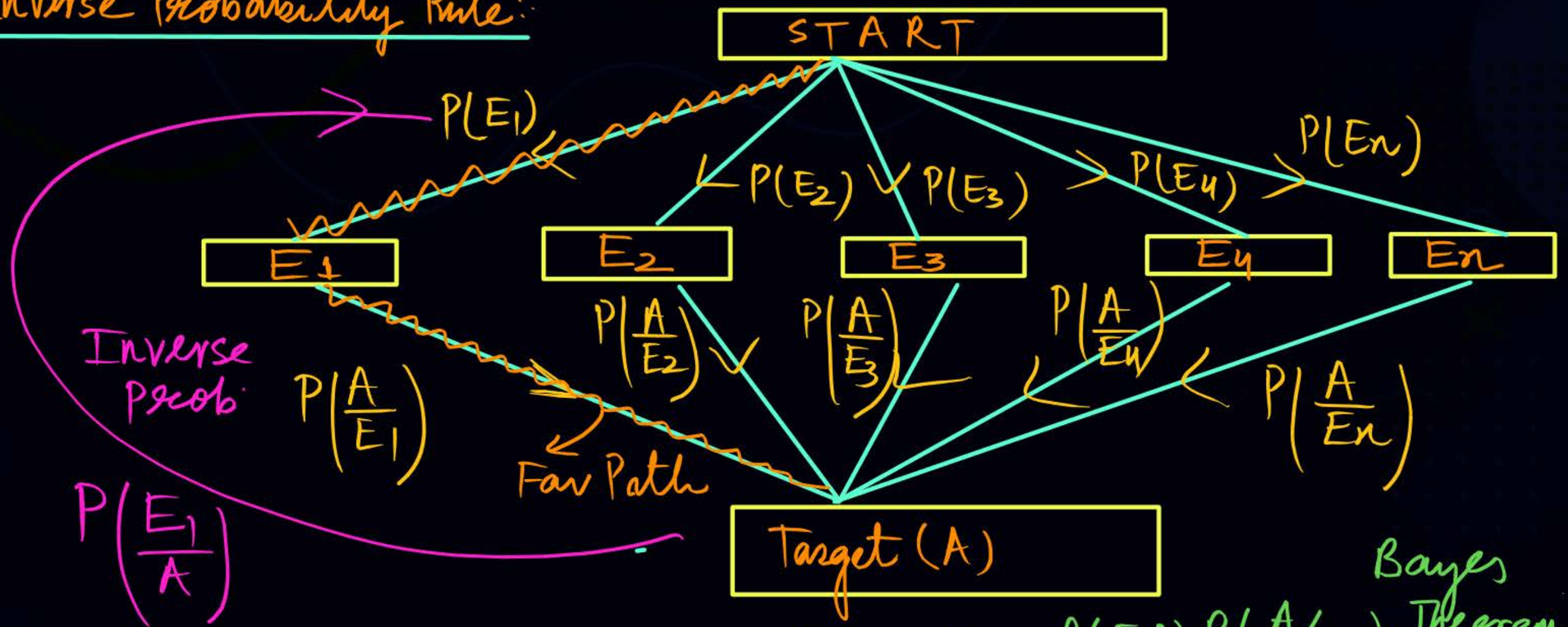


Total Prob. for Dependent events

Total Prob. Rule – Dep. events

$$P(A) = P(E_1)P\left(\frac{A}{E_1}\right) + P(E_2)P\left(\frac{A}{E_2}\right) + P(E_3)P\left(\frac{A}{E_3}\right) + - + P(E_n)P\left(\frac{A}{E_n}\right)$$

Inverse Probability Rule:



Bayes Theorem for Inverse prob

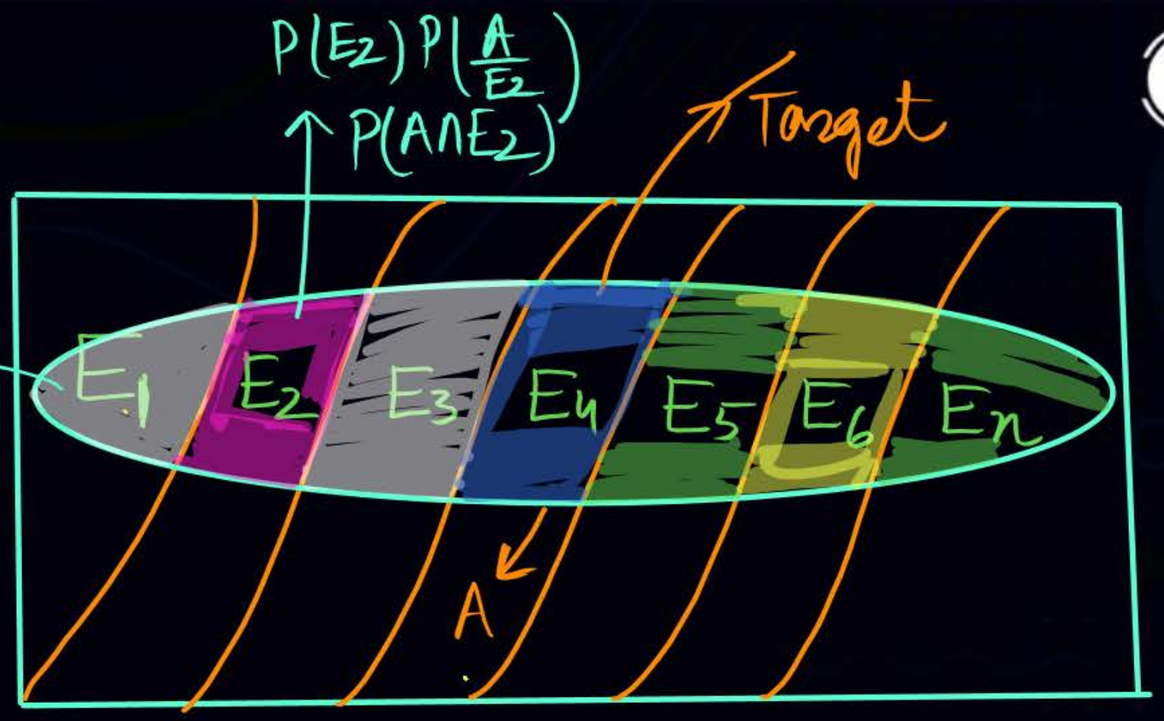
$$P\left[\frac{E_1}{A}\right] = \frac{P(E_1)\,P\left(\frac{A}{E_1}\right)}{\sum_{i=1}^{n} P(E_i)\,P\left(\frac{A}{E_i}\right)}$$

Bayes Theorem

$$P\left[\frac{E_i}{A}\right] = \frac{P(E_i)\,P\left(\frac{A}{E_i}\right)}{\sum_{i=1}^{n} P(E_i)\,P\left(\frac{A}{E_i}\right)}$$

Total prob.

For Dependent events



$$P(A \cap E_1) = P(E_1)\,P\left(\frac{A}{E_1}\right)$$

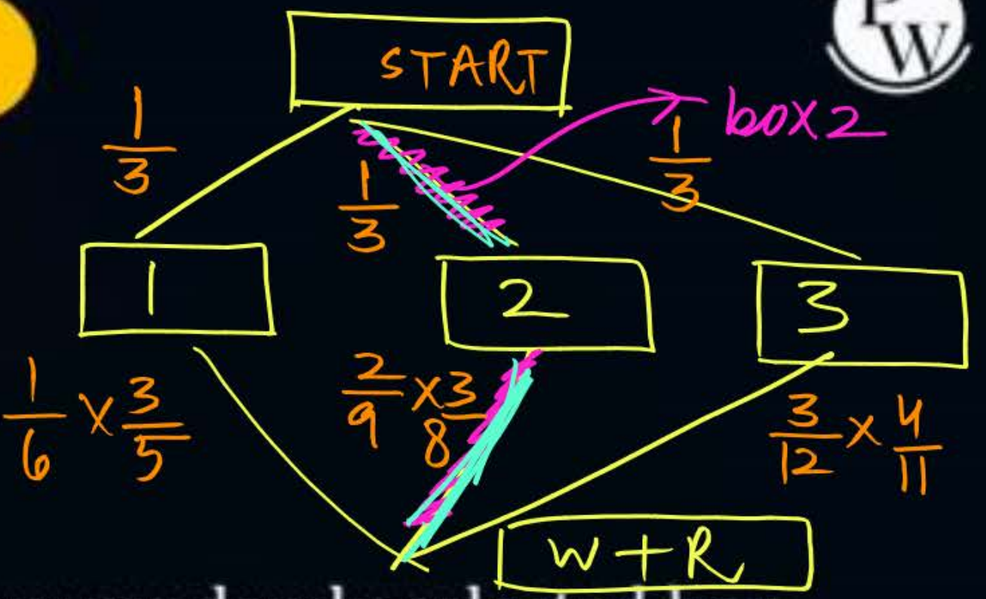
$$P(A) = P(E_1)\,P\left(\frac{A}{E_1}\right) + P(E_2)\,P\left(\frac{A}{E_2}\right) + \dots + P(E_n)\,P\left(\frac{A}{E_n}\right)$$

$$P\left(\frac{E_i}{A}\right) = \frac{P(E_i)\,P\left(\frac{A}{E_i}\right)}{\sum_{i=1}^{n} P(E_i)\,P\left(\frac{A}{E_i}\right)}$$

$i = 1, 2, 3, 4, 5 \dots$

Bayes Theorem

Topic : Probability and Statistics



#Q. Three boxes B_1, B_2, B_3 contains balls

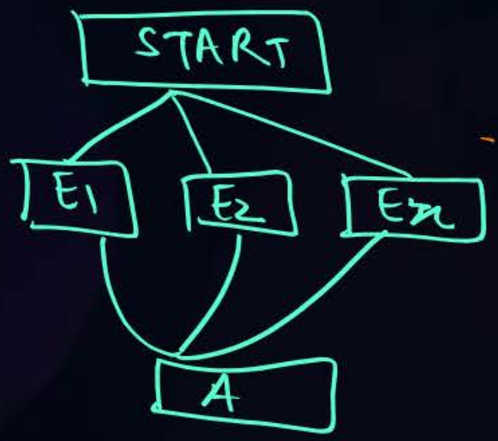
$B_1 \rightarrow$ 1W, 2B , 3R ✓ $= \frac{1}{6} \times \frac{3}{5}$

$B_2 \rightarrow$ 2W, 4B , 3R ✓ $= \frac{2}{9} \times \frac{3}{8}$

$B_3 \rightarrow$ 3W, 5B , 4R ✓ $= \frac{3}{12} \times \frac{4}{11}$

Without replacement if 2 balls are drawn from randomly selected box. Find the probability one of the ball drawn is white and other ball is red from box 2 order is specified.

W+R



$$\Rightarrow \frac{\frac{1}{3} \times \frac{2}{9} \times \frac{3}{8}}{\frac{1}{3} \times \frac{1}{6} \times \frac{3}{5} + \frac{1}{3} \times \frac{2}{9} \times \frac{3}{8} + \frac{1}{3} \times \frac{3}{12} \times \frac{4}{11}} = \frac{55}{181}$$



Topic : Probability and Statistics

#Q. One ticket is selected at random for 100 ticket 00, 01, 02,....99. Suppose A and B are the sum and product of digits found on the ticket.

Then $P\left[\frac{A=7}{B=0}\right]$ is given by

100 TKts
00, 01, 02 ... 99

D_1 D_2
SUM = A
product = B

(A) 2/13

(B) 2/19

(C) 1/50

(D) None of these

~~43 61 52~~
~~34 16 25~~

$$P\left[\frac{A=7}{B=0}\right] = P\left[\frac{sum=7}{product=0}\right] = \text{Conditional probs}$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

Using conditional Prob both occurs

$$P\left[\frac{sum=7}{pro=0}\right] = \frac{P[sum=7 \cap (product=0)]}{P[product=0]}$$

$$= \frac{n(sum=7 \cap product=0)}{\cancel{n(S)}} \Big/ \frac{n(product=0)}{\cancel{n(S)}}$$

$$= \frac{2}{19}$$

00
01
02
03
04
05
06
.
.
.
99

07, 70

00 50
01 60
02 70
03 80
04 90
05
06
07
08
09
10
20
30
40

product = 0

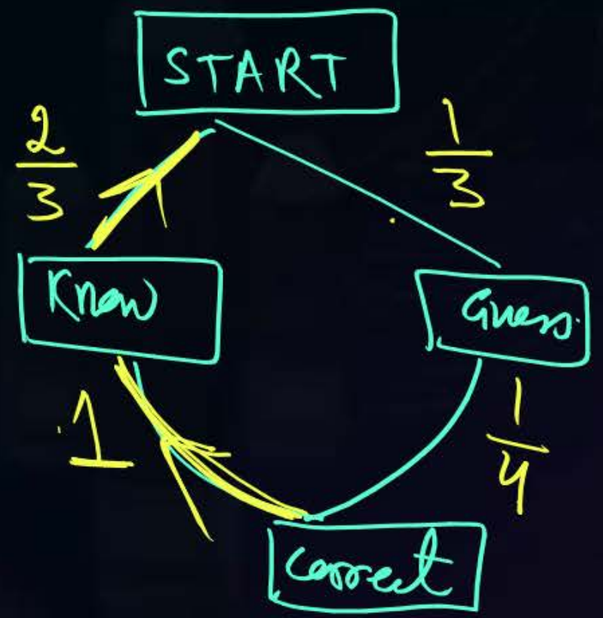
$= 19$



Topic : Probability and Statistics

$E_1, E_2, E_3 - - -$ → A depends.
E_n = mutually exclusive
A, B, C, D $= \frac{1}{4}$

#Q. The bribability that a student knows the correct answer to a multiple choice question is 2/3. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is ¼. Given that the student has answered the questions correctly, the conditional probability that the student known the correct answer is-



A 2/3

B 3/4

C 5/6

D 8/9

$= \frac{\frac{2}{3} \times 1}{\frac{2}{3} \times 1 + \frac{1}{3} \times \frac{1}{4}} = \frac{\frac{2}{3}}{\frac{27}{36}}$

$\frac{2}{3} + \frac{1}{12} = \frac{27}{36}$

$= \frac{72}{81} = \frac{8}{9}$



Topic : Probability and Statistics

1 – 100 (both included)
2 – 99 (both excluded)

#Q. The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is-

$$P(\text{Not divisible by } 2,3 \text{ or } 5) = 1 - P(\text{divisible } 2 \text{ or } 3 \text{ or } 5)$$
$$= 1 - P(A \cup B \cup C)$$

$$P(A \cup B \cup C) = \frac{74}{100} \quad \text{(do yourself)}$$

$$P(\text{Not divisible}) = 1 - \frac{74}{100}$$
$$= \frac{26}{100} = \frac{13}{50}$$

Wednesday

THANK - YOU